system we are using, the stable ratio, if you work it out, turns out to be $\frac{5}{12}$ doves to $\frac{7}{12}$ hawks. When this stable ratio is reached, the average pay-off for hawks is exactly equal to the average pay-off for doves. Therefore selection does not favour either one of them over the other. If the number of hawks in the population started to drift upwards so that the ratio was no longer $\frac{7}{12}$, doves would start to gain an extra advantage, and the ratio would swing back to the stable state. Just as we shall find the stable sex ratio to be 50:50, so the stable hawk to dove ratio in this hypothetical example is 7:5. In either case, if there are oscillations about the stable point, they need not be very large ones.

Superficially, this sounds a little like group selection, but it is really nothing of the kind. It sounds like group selection because it enables us to think of a population as having a stable equilibrium to which it tends to return when disturbed. But the ESS is a much more subtle concept than group selection. It has nothing to do with some groups being more successful than others. This can be nicely illustrated using the arbitrary points system of our hypothetical example. The average pay-off to an individual in a stable population consisting of $\frac{7}{12}$ hawks and $\frac{5}{12}$ doves turns out to be $6\frac{1}{4}$. This is true whether the individual is a hawk or a dove. Now $6\frac{1}{4}$ is much less than the average pay-off for a dove in a population of doves (15). If *only* everybody would agree to be a dove, every single individual would benefit. By simple group selection, any group in which all individuals mutually agree to be doves would be far more successful than a rival group sitting at the ESS ratio. (As a matter of fact, a conspiracy of nothing but doves is not quite the most successful possible group. In a group consisting of $\frac{1}{6}$ hawks and $\frac{5}{6}$ doves, the average pay-off per contest is $16\frac{2}{3}$. This is the most successful possible conspiracy, but for present purposes we can ignore it. A simpler all-dove conspiracy, with its average pay-off for each individual of 15, is far better for every single individual than the ESS would be.) Group selection theory would therefore predict a tendency to evolve towards an